

Supplementary Materials

Full Survey Measures & Description of all survey measures in order.....	2
Additional Analyses: Group Attitudes (Excluding Gender and Religion Items).....	5
Additional Analyses: Social Identity Importance (Excluding Gender and Religion Items)	8
Additional Analyses: Perception of School Norms.....	11
Additional Analyses: Personal Well-Being Items.....	13
Additional Analyses: Religious Observance Items	14
Additional Analyses: Socioeconomic status items.....	17
Additional Analyses: Resource Allocation.	18
Model Construction: Variable Construction	20
Model Construction: General Model Specification and Selection Procedure.....	21
Model Construction: Base Models.....	23
Grade-level overview of graph-level network statistics for positive and negative nominations.	28
Trends across grade in four common graph-level networks statistics: betweenness, transitivity, reciprocity, and network density.	29
Social Network Analysis of Those who Engaged in Cross-Group Friendships	30

Online Supplemental Material 1

Full Survey Measures & Description of all survey measures in order

The complete survey measures are provided in this repository as “social nets survey older kids.pdf” and “social nets survey younger kids.pdf”. Below is a detailed description of all survey measures in order they were administered. Unless otherwise indicated, items were identical across survey versions. A longer version was administered to students in Grade 5 and above, whereas a shorter version was administered to students in Grade 3 and 4.

- 1) *Positive ties* (friendship) in which students were asked to select up to seven individuals they considered their “closest friends in (their) class” from a roster of all students in their classroom. Despite this instruction, in some cases students selected more than 7 friends; since there was no way to determine the order in which friends were selected, we included all selected friends for those children.
- 2) *Negative ties*, in which students were asked to select up to seven individuals they “don’t spend time with or that (they) try to avoid” from another copy of the same class roster. If more than 7 were selected, we again included all selected negative ties. These items were not included in the short version administered to younger children.
- 3) *Group attitudes*, each involving a single 7-point Likert-like item indicating liking from “Dislike very much” to “Like very much”. There were 12 items: Hindus, Muslims, rich people, poor people, people who speak the child’s same language, people who speak a different language, people of the child’s caste, people of other castes, children who get good marks (grades), children who get bad marks, boys, and girls. The shorter version omitted the items targeting good and bad marks.

- 4) *Social identity importance*, each involving a single 5-point Likert-like item indicating importance from “Very important” to “not at all important”, each for different dimensions of social identity, namely the importance to the child of the school, their religion, their gender, their language, their caste, and their marks (grades) in school.
- 5) *Perception of school norms of intergroup interaction*, each involving a single 5-point Likert-like item indicating agreement from “definitely agree” to “definitely disagree”; the norms were that most children at the school get along, that it’s okay for boys and girls to be friends, that it’s okay for Hindu and Muslim children to be friends, that it’s okay for children of different castes to get along, and that it’s okay for richer and poorer children to be friends. These items were not included in the short version administered to younger children.
- 6) *Personal social well-being questions*, each involving a single 5-point Likert-like item indicating agreement from “definitely agree” to “definitely disagree”; the items were whether the child felt they had enough friends, whether other children were sometimes mean to the child, whether they felt lonely, and whether they were happy overall.
- 7) *Religiosity*, in which children rated themselves on a single four-point item rating of own religiosity ranging from “very religious” to “not at all religious.”
- 8) *Questions about religious observance*, including four five-point items on frequency of religious observance ranging from “every day” to “never/almost never” for the following religious activities: Pray to god; go to mosque, temple, or church; celebrate a religious holiday or event; read holy books.
- 9) *Questions about socioeconomic status*, including the MacArthur Ladder, in which children position their family on a 10-rung ladder ranging from the bottom economic

rung to the top by comparing their family to the set of people in their city; a question about the number of working adults in the home; a question about how often the child gets spending money on a 4 point scale ranging from “every day” to “less than every month”; and a question about whether the child worries about how much money the family has on a 4 point scale, ranging from “very often” to “very rarely”. The short version administered to younger children included only the spending money item of this set; the MacArthur Ladder and the item about worrying about money were not included.

10) *Hypothetical resource allocation behavior* in the form of three hypothetical dictator games in which the child is asked how many candies they would give to another child without candies if they themselves had 6 sweets (i.e., candies) (ranging from 0 to 6) for an unnamed anonymous child, a stranger child with a characteristic Muslim name (Mohammed), and a child with a characteristic Hindu name (Krishna).

11) *Demographic items*, namely religion, gender, caste, and language most often spoken at home. Age was computed from their date of birth and so was not solicited in the survey.

The shorter version of the survey administered to students in Grades 3 and 4 retained the same general order but omitted the following:

- 1) The negative ties measure
- 2) The group-attitude items concerning children who receive good and bad grades.
- 3) The school norm items
- 4) The MacArthur Ladder and the question about how often they worry about their family’s money.

Online Supplemental Material 2

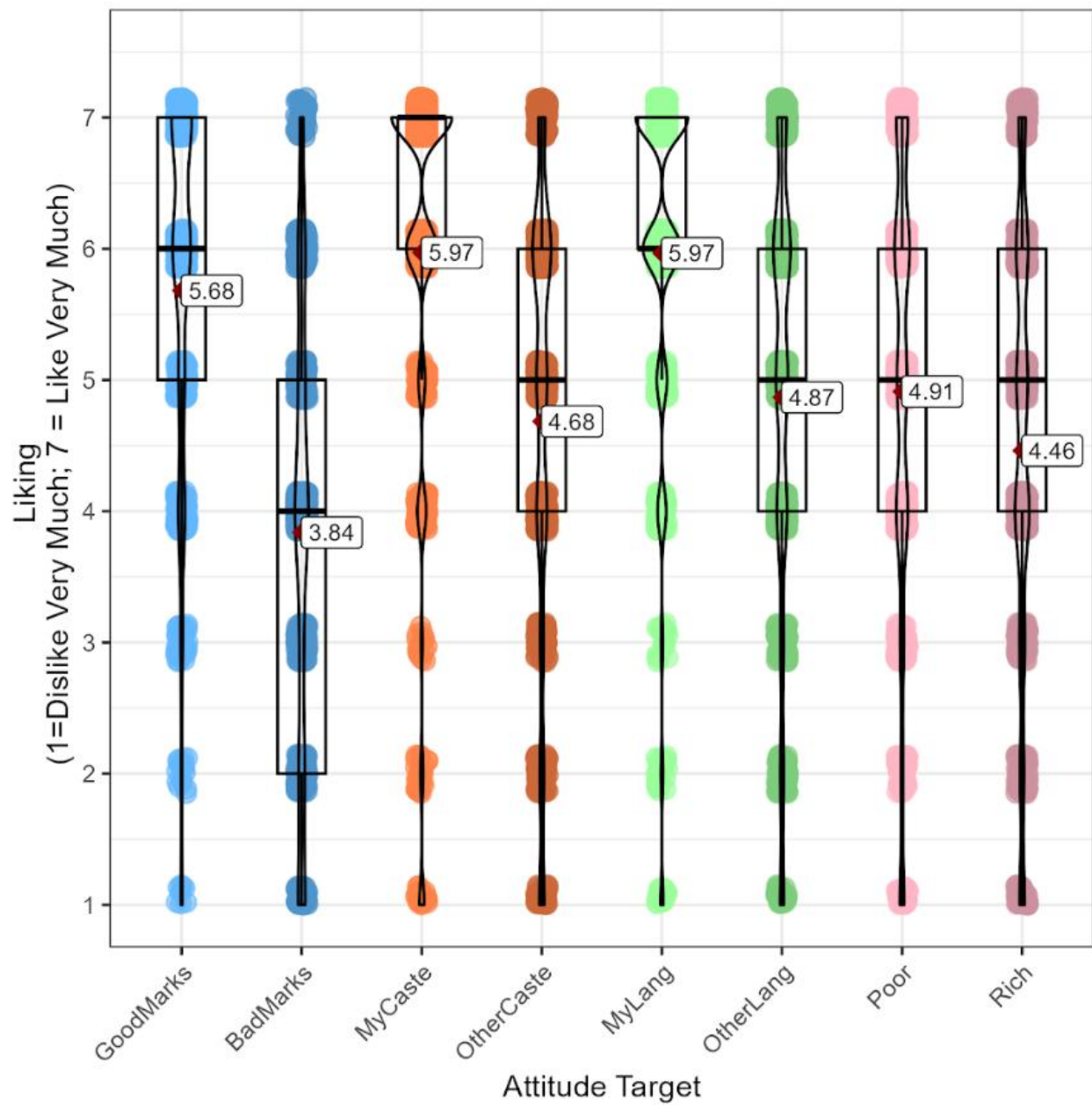
Additional Analyses: Group Attitudes (Excluding Gender and Religion Items)

This section summarizes responses to the group-attitude items unrelated to gender or religion; these items are reported in the main manuscript. We first report general trends and then discuss whether there were any gender, religion, or age-related changes in group attitudes.

Figure S1 summarizes overall means for the 8 other attitude items, which generally revealed ingroup or high-status preference, with robust preferences for children with good over bad marks, own versus other caste, and own versus other language. Outgroup attitudes were not particularly negative, generally averaging near the midpoint of the scale. The one exception was wealth attitudes, in which children actually reported somewhat more positive attitudes towards poor than rich children.

Figure S1

Ratings of Ingroup and Outgroup Caste and Language Targets and other Social Categories



Note: Each dot reflects a participant response to that question, jittered to avoid over-plotting, with overlaid box and violin plots.

We next examined potential gender and religion differences in these attitude items, using a corrected p -value of $0.05/10 = 0.005$ tests for multiple comparisons. Gender differences

appeared in only two of the non-gender attitude items, with girls showing more positive attitudes towards speakers of their own language ($M_{girls} = 6.1$, $M_{boys} = 5.9$, $t(1178) = 2.99$, $p = .003$, $d = .17$) and towards people with good marks ($M_{girls} = 5.8$, $M_{boys} = 5.6$, $t(880) = 2.88$, $p = .004$, $d = .20$). Religious differences appeared for attitudes towards speakers of one's own language ($M_{Hindu} = 6.1$, $M_{Muslim} = 5.8$, $t(1003) = 3.93$, $p < .001$, $d = .23$), those with good marks ($M_{Hindu} = 5.8$, $M_{Muslim} = 5.5$, $t(749) = 3.05$, $p = .002$, $d = .21$), and wealthy individuals ($M_{Hindu} = 4.6$, $M_{Muslim} = 4.2$, $t(1082) = 3.50$, $p < .001$, $d = .21$).

Attitudes also exhibited age-related change, generally growing more positive with age (other caste: $r = .24$; bad marks: $r = .28$; other language: $r = .28$; the poor: $r = .12$; own caste: $r = .11$; all $p < .001$), with three exceptions: attitudes towards those with good marks, which trended downward ($r = -.16$, $p < .001$), and attitudes towards one's own language and towards rich people, which were generally stable ($|r| < .05$, $p > .13$).

Online Supplemental Material 3

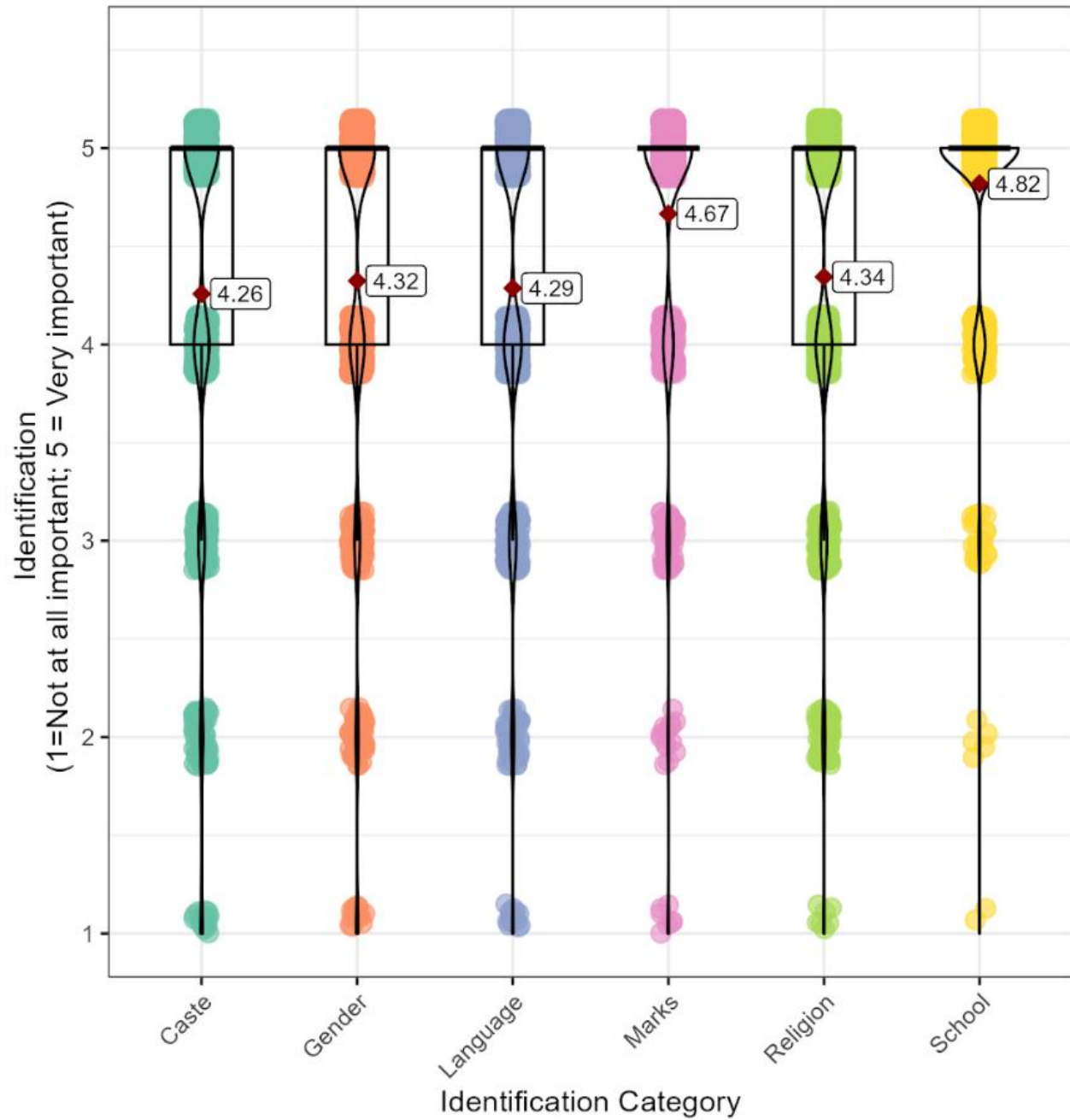
Additional Analyses: Social Identity Importance (Excluding Gender and Religion Items)

This section summarizes responses to the social identity importance items unrelated to gender or religion; these items are reported in the main manuscript. We first report general trends and then discuss whether there were any gender, religion, or age-related changes in social identity importance.

Figure S2 demonstrates the levels of identification across social groups. Identification was quite high across all target groups, with means above 4 on the 5-point scale. Identification with academic achievement and with school stood out as somewhat higher than the others, though all were above the scale midpoint.

Figure S2

Identification with Various Social Identities



Note: Each dot reflects a participant response to that question, jittered to avoid over-plotting, with overlaid box and violin plots.

We next examined potential religious or gender-related differences in social identity importance, using a corrected p -value of $0.05/6 = 0.0083$ tests for multiple comparisons. No

gender differences emerged using this threshold. However, a number of religion differences between Hindus and Muslims did emerge. Surprisingly given the seemingly more central role of a caste system for Hindu children, Muslims rated caste as more important to their identity than did Hindus, ($M_{Hindu} = 4.1$, $M_{Muslim} = 4.5$, $t(1190) = 5.63$, $p < .001$, $d = .33$). Muslims also rated gender as more important to their social identity ($M_{Hindu} = 4.4$, $M_{Muslim} = 4.2$, $t(1168) = 3.46$, $p < .001$, $d = .20$) and religious identification ($M_{Hindu} = 4.2$, $M_{Muslim} = 4.6$, $t(1203) = 7.23$, $p < .001$, $d = .42$). By contrast, Hindu children rated themselves higher on school identification identified more strongly with school, ($M_{Hindu} = 4.9$, $M_{Muslim} = 4.7$, $t(848) = 4.50$, $p < .001$, $d = .26$). There were also statistically significant if modest differences with respect to participant age on several items, with gender identification increasing somewhat ($r = .13$, $p < .001$) and language ($r = -.18$), marks ($r = -.12$), and school ($r = -.14$) identification declining (all $p < .001$).

Online Supplemental Material 4

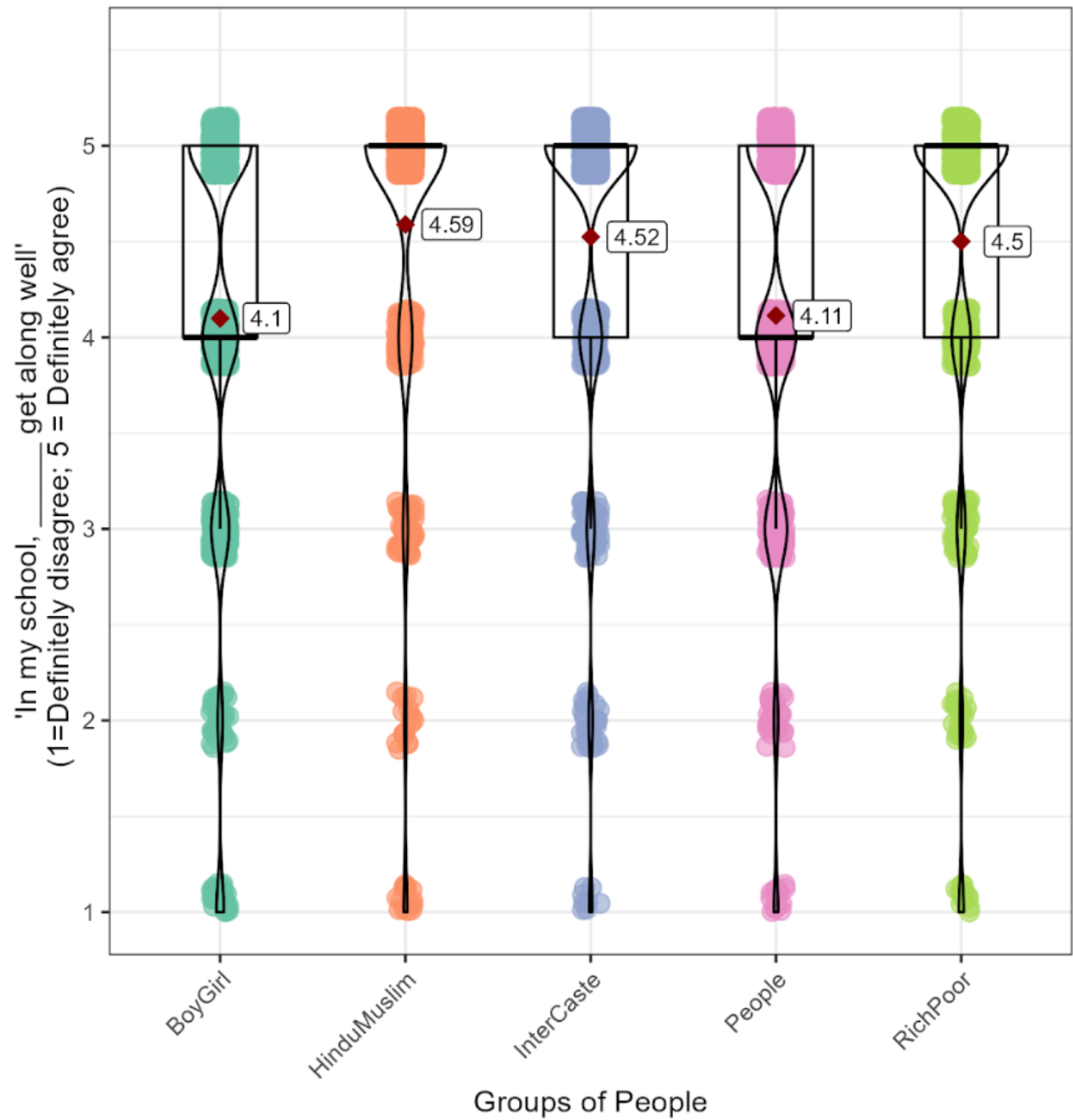
Additional Analyses: Perception of School Norms

This section summarizes responses to the additional school norm items which assessed perceptions of inter-caste relations and rich-poor relations within the school. We first report general trends and then discuss whether there were any gender, religion, or age-related changes in perceptions of school norms.

Figure S3 demonstrates the mean level of endorsement of various positive intergroup norms within the school. Children generally endorsed positive norms at their school, with all means above 4 on the 5-point scale, though positive norms concerning most people getting along and inter-gender interactions were somewhat lower than inter-caste, inter-religion, and inter-wealth interaction norms. Using a corrected threshold for multiple comparison ($p = 0.05/5 = 0.01$), none of the norms related to inter-caste or inter-wealth relations differed as a function of participant gender or religion. There were, however, pronounced age differences, with endorsement of positive norms tending to increase with age (caste norms: $r = .24$; wealth norms: $r = .16$, all $p < .001$).

Figure S3

Perception of intergroup norms within school



Note: Each dot reflects a participant response to that question, jittered to avoid over-plotting, with overlaid box and violin plots.

Online Supplemental Material 5

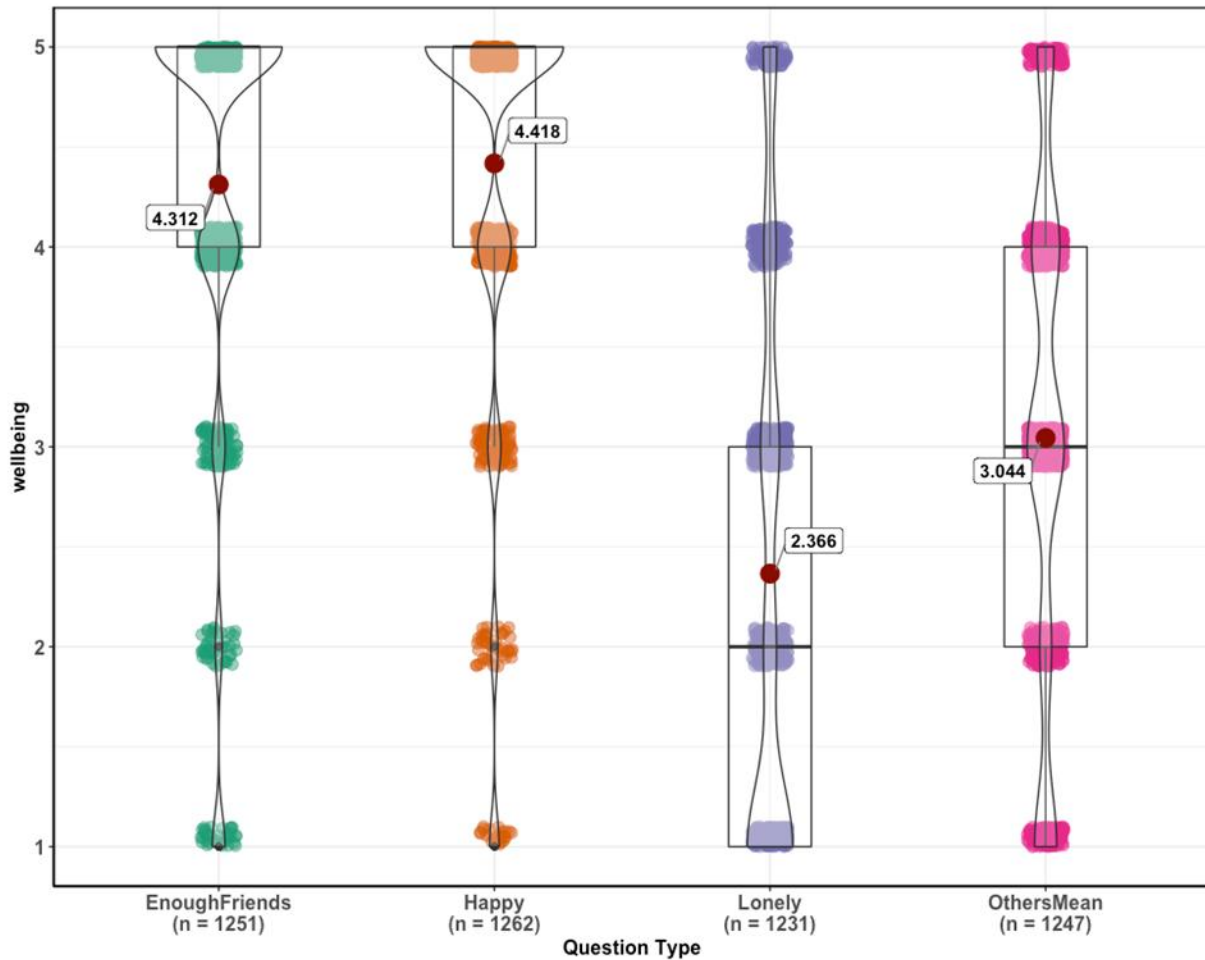
Additional Analyses: Personal Well-Being Items

This section summarizes responses to the personal well-being items. We first report general trends and then discuss whether there were any gender, religion, or age-related changes in perceptions of school norms.

Figure S4 demonstrates the mean level of endorsement related to personal-wellbeing items. Participants generally indicated that they had enough friends, were happy, and that they were not lonely. They were close to the scale mid-point for the item on whether others were sometimes mean to them. Endorsement of these items did not appear to meaningfully vary by participant gender or religion using a corrected threshold for multiple comparisons, $p = (.05/4) = .0125$. However, age differences were observed, with older children generally reporting higher happiness ($r = .19, p < .001$), lower loneliness ($r = .24, p < .001$), but somewhat higher reports of others being mean to them ($r = .12, p < .001$).

Figure S4

Ratings of Personal Wellbeing



Note: Each dot reflects a participant response to that question, jittered to avoid over-plotting, with overlaid box and violin plots.

Online Supplemental Material 6

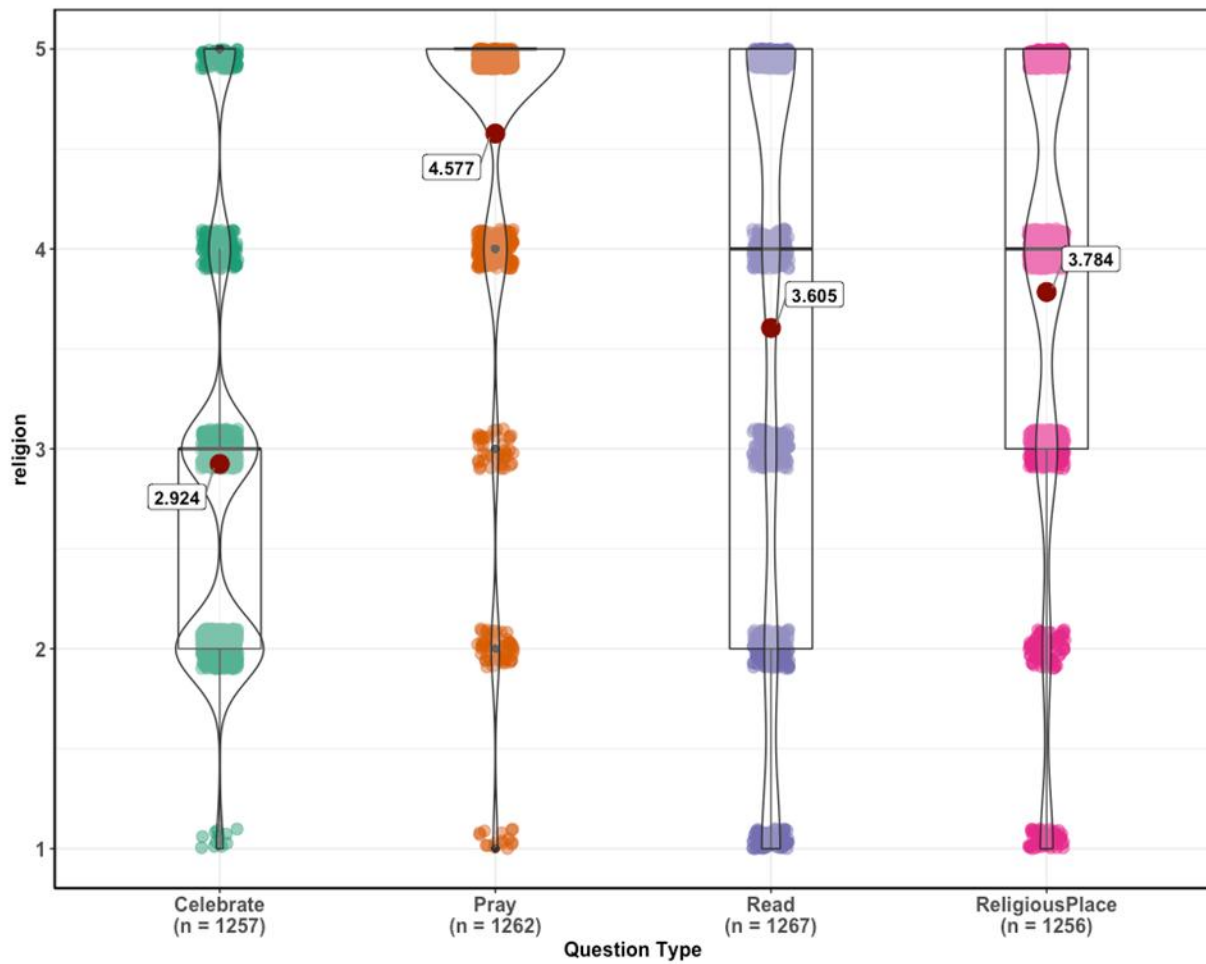
Additional Analyses: Religious Observance Items

This section summarizes responses to religious observance items. We first report general trends and then discuss whether there were any gender, religion, or age-related changes in perceptions of school norms.

Figure S5 demonstrates the mean level of endorsement of religious observance items. Levels of general religiosity were high, $M = 3.6$ ($SD = .7$) out of 4, and were somewhat higher among Muslim children, ($M_{Hindu} = 3.5$, $M_{Muslim} = 3.7$, $t(1204) = 4.39$, $p < .001$, $d = .25$). Rates of religious observance varied somewhat across items; however, caution should be taken given that frequency scales such as these may not fully meet assumptions of continuity. Nonetheless, we examined potential gender and religious differences using a corrected threshold for multiple comparisons ($p = 0.05/4 = 0.0125$). One gender difference was evident; boys reported more visits to religious places than girls (e.g. mosques or temples), ($M_{girls} = 3.6$, $M_{boys} = 3.9$, $t(983) = 4.52$, $p < .001$, $d = .26$). Two religion differences were also apparent; Muslim children reported more visits to religious places ($M_{Hindu} = 3.7$, $M_{Muslim} = 3.9$, $t(1006) = 3.96$, $p < .001$, $d = .23$) and more reading of religious books ($M_{Hindu} = 3.0$, $M_{Muslim} = 4.5$, $t(1214) = 20.04$, $p < .001$, $d = 1.16$). There were also some age differences, with older children generally reporting somewhat less overall religiosity, $r(1187) = -.33$, $p < .001$, somewhat less frequency of reading religious books, $r(1196) = -.11$, $p < .001$, but somewhat higher frequency of praying, $r(1189) = .12$, $p < .001$.

Figure S5

Religious observance



Note: Each dot reflects a participant response to that question, jittered to avoid over-plotting, with overlaid box and violin plots.

Online Supplemental Material 7

Additional Analyses: Socioeconomic status items.

This section summarizes responses to the socioeconomic status items. We first report general trends and then discuss whether there were any gender, religion, or age-related changes in the socioeconomic status items.

The modal response for number of working adults in the home was 1 (356, 40% of respondents), though responses varied, with a number of children reporting 2 (221 = 25%), 3 (127 = 14%), 4 (86 = 10%), or none (91 = 10%), with 19 (2%) not providing a response. On the MacArthur Ladder the mean response (with 1 scored as the highest and 10 as the lowest) was 4.1 ($SD = 1.7$). The modal response to the question about frequency of worrying about money was “somewhat often,” corresponding to 2 on a 4-point scale from “very often” to “very rarely”, which was chosen by 350 (39%) though all responses appeared frequently in the data (very often: 213 = 24%; rarely: 203 = 23%; very rarely: 132 = 15%, did not respond: 2 = .02%). For the final question about frequency of receiving pocket money, the modal response was “every month”, chosen by 265 respondents (29%), though again responses were well distributed (every day: 240 = 27%; every week: 231 = 26%; less than once a month: 162 = 18%, did not respond: 2 = .22%). No gender or age differences emerged on these items, but for the last two frequency items our data suggested lower economic status for Muslim children, who reported somewhat more worry about money ($M_{Hindu} = 2.43$, $M_{Muslim} = 2.2$, $t(822) = 2.47$, $p = .01$, $d = .17$; though when analyzed non-parametrically the difference was only marginal, $\chi^2(3) = 6.36$, $p = .095$) and who reported receiving pocket money quite a bit less often ($M_{Hindu} = 2.67$, $M_{Muslim} = 2.0$, $t(780) = 9.56$, $p < .001$, $d = .69$; non-parametrically, $\chi^2(3) = 106.1$, $p < .001$).

Data for these items are available from the authors upon reasonable request.

Online Supplemental Material 8

Additional Analyses: Resource Allocation.

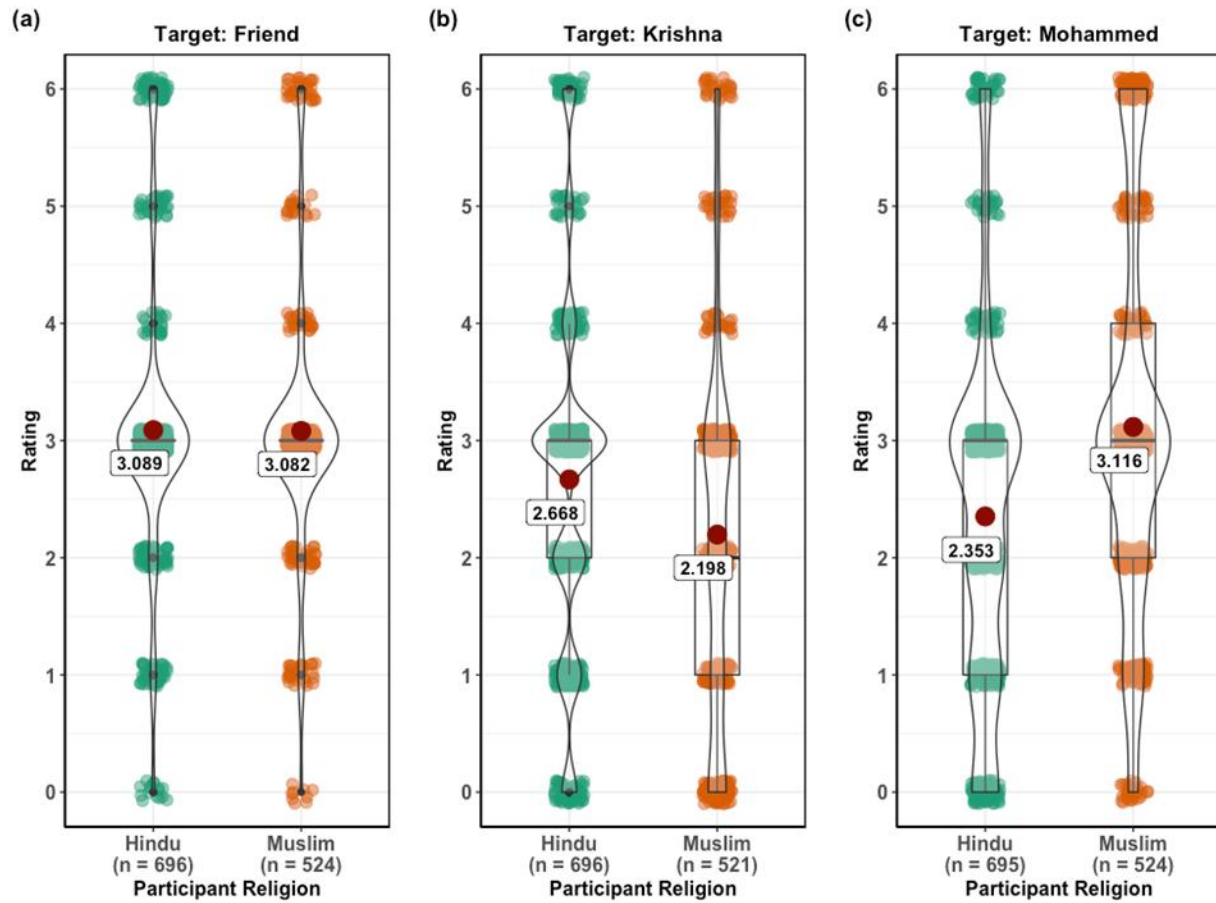
This section summarizes responses to the resource allocation items. We first report general trends and then discuss whether there were any gender, religion, or age-related changes in the resource allocation.

Figure S6 demonstrates the mean level of sweets (i.e., candies) participants decided to give strangers with no sweets of different religions, separated by religion. Resource allocation decisions showed clear evidence of own-religion favoritism and so we present results separately for Muslim and Hindu children in Figure 8. While there were no differences in terms of giving to a friend ($M_{Hindu} = 3.1$, $M_{Muslim} = 3.1$, $t(1161) = .11$, $p = .91$), own-religion favoritism appeared on the two religiously-tinged items, with Hindus giving more to Krishna ($M_{Hindu} = 2.7$, $M_{Muslim} = 2.2$, $t(1059) = 5.14$, $p < .001$, $d = .32$) and Muslims giving more to Mohammed ($M_{Hindu} = 2.4$, $M_{Muslim} = 3.1$, $t(1079) = 8.25$, $p < .001$, $d = .50$). No gender or age differences met our corrected threshold ($p = .05 / 3 = .016$). Thus, children generally indicated a desire to donate about half their hypothetical endowment to friends and religious ingroup members, and somewhat less to religious outgroup members.

Figure S6:

Responses to hypothetical resource allocation (dictator game) task.

Rating of gender targets based on participant gender



Note: Each dot reflects a participant response to that question, jittered to avoid over-plotting, with overlaid box and violin plots, grouped by participant religion (x-axis) and faceted by target of the resource allocation.

Online Supplemental Material 9

Model Construction: Variable Construction

Peer nominations were coded as binary outcomes indicating whether a nomination was same-group (1) or cross-group (0), separately for gender- and religion-based networks and for positive and negative nomination types.

Classroom composition variables were computed as proportions within each classroom, including proportion male, proportion Hindu, proportion other religion (non-Hindu or Muslim), classroom size, and internal edge density.

Centered grade was computed by subtracting the sample mean grade from each participant's grade. Own-group preference for gender and religion was operationalized using difference scores, calculated as ingroup liking minus outgroup liking. Own-group preference, social identification, and perception of group norms were measured at the individual level and entered as continuous predictors.

Online Supplemental Material 10

Model Construction: General Model Specification and Selection Procedure

We estimated multilevel models with random intercepts for students and classrooms. All models were initially specified as maximal models including:

- Participant gender,
- Participant religion
- Centered grade
- Classroom composition variables
 - Proportion of Males
 - Proportion of Hindus
 - Proportion of Other religions (non-Hindu or Muslim)
 - Class Size
 - Internal Edge Density
- All theoretically relevant two-way and higher order interactions

Model simplification proceeded using a backward elimination procedure based on likelihood ratio tests (LRTs). Beginning with highest-order interaction terms, non-significant effects ($p > .05$) were sequentially removed. At each step model comparisons were conducted using nested likelihood ratio tests. Lower-order terms were retained when involved in significant higher-order interactions. Classroom composition variables were retained across all models regardless of statistical significance due to their role in capturing baseline nominations structure. On average, 12-18 interaction terms were evaluated per model sequence. Full model comparison sequences (including all intermediate steps and LRT statistics) are available upon request.

To evaluate whether additional factors (e.g., ingroup preference) explained variance beyond the base models, we estimated a series of models in which each predictor was added separately to the relevant base specification. Predictors included: intergroup attitudes, social identification variables, and perceptions of school norms. For each construct, we tested main effects, interactions with the relevant participant dimensions (e.g., participant gender for gender outcomes), and interactions with centered grade. Non-significant interaction terms were removed using likelihood ratio tests consistent with the base modeling strategy. These models were not combined into a single selection procedure; rather, they were used to assess incremental explanatory value relative to the base model.

Online Supplemental Material 11*Model Construction: Base Models*

Final reduced base models are reported in the main text. For transparency, we additionally provide full model summaries for each final specification:

Table S1: Positive Gender Nominations

Step	Term evaluated	Action	Criterion	Result
1	Full interaction model specified	—	—	Baseline model
2-6	Three way interactions	LRT	$p > .06$	removed
7-14	Two way interactions	LRT	$p > .10$	remove
15	PropHindu	LRT	$p = .89$	removed
16	PropOther	LRT	$p = .50$	removed
17	PropMale	LRT	$p = .37$	Not removed because accounts for base rates
Base	ownGenderNomination ~ Gender + Grade + propMale + ClassSize + InternalEdge + (1 Subject) + (1 Class)			

Table S2: Positive Religion Nominations

Step	Term evaluated	Action	Criterion	Result
1	Full intaction model specified	—	—	Baseline model
2-5	Three way interactions	LRT	$p > .06$	removed
6	Grade:ClassSize	LRT	$p = .74$	removed
7	Religion:ClassSize	LRT	$p = .68$	removed
8	Grade:PropOther	LRT	$p = .53$	removed
9	Grade:PropHindu	LRT	$p = .53$	removed
10	Religion:PropMale	LRT	$p = .50$	removed
11	Religion:InternalEdge	LRT	$p = .24$	removed
12	Grade:PropMale	LRT	$p = .21$	removed
13	Grade:InternalEdge	LRT	$p = .08$	removed
14	InternalEdge	LRT	$p = .99$	removed
15	ClassSize	LRT	$p = .83$	removed
Base	OwnReligionNomination ~ Religion + Grade + PropMale + PropHindu + PropOther + Religion*PropHindu + Religion*PropOther + (1 Subject) + (1 Class)			

Table S3: Negative Gender Nominations

Step	Term evaluated	Action	Criterion	Result
1	Full intaction model specified	—	—	Baseline model
2-6	Three way interactions	LRT	$p > .50$	removed one by one
7	Grade:ClassSize	LRT	$p = .97$	removed
15	All non-significant 2 ways	LRT	$p > .30$	Removed all, Retain Gender:PropOther and Gender:Grade
Base	ownGenderNegNomination ~ Gender + Grade + PropMale + PropOther + InternalEdge + Gender*PropOther + Gender*Grade + (1 Subject) + (1 Class)			

Table S4: Negative Religion Nominations

Step	Term evaluated	Action	Criterion	Result
1	Full intaction model specified	—	—	Baseline model
2-6	All non-significant 3-ways	LRT	$p > .18$	Removed
11	All non-significant 2-ways	LRT	$p > .20$	Removed Retain Religion * PropMale, Religion * PropHindu, Religion * PropOther, Religion * ClassSize, Religion * InternalEdge
17	PropMale	LRT	$p = .37$	Not removed because accounts for base rates
Base	ownReligionNomination ~ Religion + Grade + PropMale + PropHindu + PropOther + ClassSize + InternalEdge + Religion * PropHindu + Religion * PropOther + Religion * ClassSize + Religion * InternalEdge + (1 Subject) + (1 Class)			

Online Supplemental Material 12

Grade-level overview of graph-level network statistics for positive and negative nominations.

Table S5

Grade	Positive Ties (Friendships)								
	Average Degree	SD of Degree	In-degree SD	Out-degree SD	Density	Transitivity	Betweenness	Avg Path Length	Reciprocity
3	10.948	4.02	3.133	2.196	0.078	0.306	140.019	3.178	0.419
4	10.981	4.225	3.367	2.197	0.086	0.357	140.067	3.37	0.458
5	11.363	3.844	2.768	2.194	0.091	0.351	119.677	3.231	0.489
6	11.253	4.268	2.733	2.715	0.089	0.41	89.081	2.983	0.457
7	11.651	4.141	3.249	2.067	0.091	0.367	123.723	3.131	0.489
8	9.351	3.734	2.404	2.991	0.071	0.336	110.601	3.633	0.396
9	12.558	4.199	3.292	2.202	0.096	0.348	112.846	2.875	0.488
10	11.474	4.175	2.656	2.703	0.102	0.396	97.07	3.102	0.575

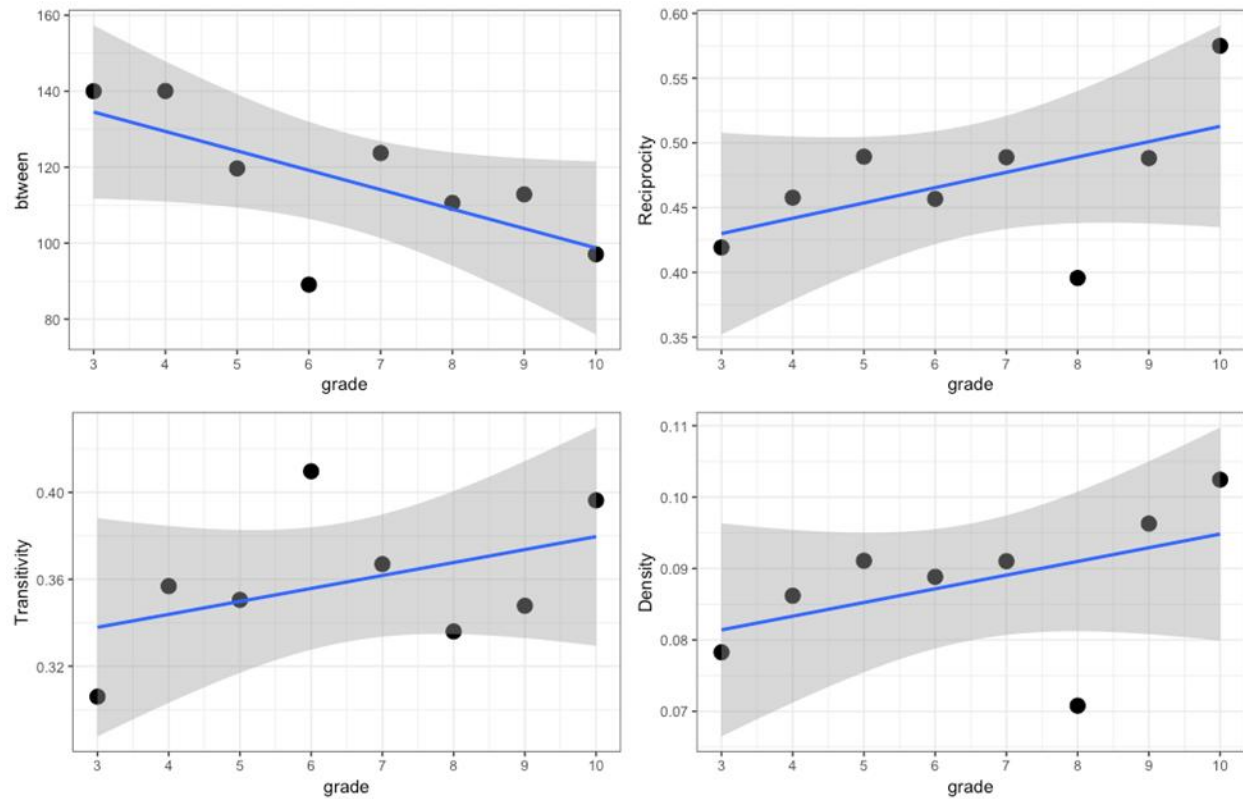
Table S6

Negative Ties (Avoidance)									
Grade	Average Degree	SD of Degree	In-degree SD	Out-degree SD	Density	Transitivity	Betweenness	Avg Path Length	Reciprocity
5	6.899	4.355	3.422	3.113	0.059	0.149	80.821	3.404	0.123
6	7.265	4.226	3.198	3.117	0.059	0.175	105.577	3.777	0.103
7	4.077	3.281	2.055	2.574	0.046	0.118	24.75	2.969	0.055
8	5.276	3.341	2.32	2.849	0.043	0.165	32.974	3.021	0.052
9	6.709	4.399	3.409	2.863	0.054	0.149	81.916	3.312	0.073
10	4	2.92	1.864	2.763	0.053	0.074	10.846	2.516	0.051

Note for Table S5 & S6: Grade-level values are simple averages of the three classrooms making up that grade (except for Grade 10, in which there is just one classroom in the dataset). Grades 3 and 4 were not asked about negative ties.

Online Supplemental Material 13

Trends across grade in four common graph-level networks statistics: betweenness, transitivity, reciprocity, and network density.



Online Supplemental Material 14

Social Network Analysis of Those who Engaged in Cross-Group Friendships

To explore the implications of frequently versus rarely forming friendships across group lines, we designated subsets of the children in our sample who more often versus less often (or never) crossed group boundaries. “High religion crossers” were designated as children whose proportion of friendships with other-religion individuals was at or above the 75th percentile (≥ 1.0 ; $N_{\text{high}}=318$ out of a total $N = 1234$). This group consisted of children whose friendship networks had equal to or above 43% other-religion individuals. Because cross-gender friendships were substantially less common (Figure 4), “high gender crossers” were designated as children who had at least one other-gender friend ($N_{\text{high}}=207$ out of a total $N = 1234$). Consistent with the findings of our logistic regression models (see Part 3 of Results), girls were more likely than boys to be classified as high religion crossers (31.7% vs. 21.9%), $\chi^2(1) = 14.37, p < .001$, and Muslim children were more likely than Hindu children to be high religion crossers (34.8% vs. 18.9%), $\chi^2(1) = 39.26, p < .001$. Further, girls were also more likely than boys to be classified as high gender crossers (21.5% vs. 13.7%), $\chi^2(1) = 12.25, p < .001$, and Hindu children were more likely than Muslim children to be high gender crossers (19.1% vs. 13.7%), $\chi^2(1) = 5.74, p = .017$. High gender crossers were also more likely to be in higher grades ($M = 6.43$ vs. 5.97), $t(272.89) = -2.55, p = .011$. There was no significant association between being a high religion crosser and a high gender crosser, $\chi^2(1) = 0.25, p = .62$.

Outgoing Nominations. The number of peers a child nominates as friends can signify how socially outgoing vs. withdrawn they are (Clifton et al., 2009). Consistent with the idea that children who frequently crossed group lines were more outgoing, high gender crossers made significantly more positive nominations than other children ($M_{\text{high}} = 6.35, M_{\text{other}} = 5.56$), t

(290.44) = -6.17, $p < .001$, 95% CI [-1.04, -0.54], and this effect was significant among high religion crossers as well ($M_{\text{high}} = 5.85$, $M_{\text{other}} = 5.64$), $t(636.68) = -2.06$, $p = .039$, 95% CI [-0.41, -0.01].

Indegree Nominations. The number of times a child was nominated by other children in their social network could signify their social popularity (Conti et al., 2013). Interestingly, while high gender crossers were nominated as friends significantly more than others ($M_{\text{high}} = 6.14$, $M_{\text{other}} = 5.64$), $t(279.45) = -2.00$, $p = .047$, 95% CI [-1.00, -0.01], high religion crossers were nominated marginally less than others ($M_{\text{high}} = 5.45$, $M_{\text{other}} = 5.82$), $t(535.14) = 1.77$, $p = .08$, 95% CI [-0.04, 0.77].

Closeness. Closeness reflects how close an individual is to all others in the network, defined as the inverse of the average shortest path length from that individual to all others. For example, if a child has an average shortest path length of 2 to all other peers, their closeness value would be $\frac{1}{2}$. High closeness indicates that an individual occupies a more central position in the network, whereas low closeness suggests that it takes many steps to reach others, potentially indicating greater social isolation (Prochnow, 2025). High gender crossers were significantly higher in closeness than others ($M_{\text{high}} = 0.45$, $M_{\text{other}} = 0.39$), $t(348.08) = -18.41$, $p < .001$, 95% CI [-0.07, -0.06], while high religion crossers were not significantly higher or lower in closeness than others ($M_{\text{high}} = 0.40$, $M_{\text{other}} = 0.40$), $t(586.29) = -1.05$, $p = .294$, 95% CI [-0.01, 0.00].

Betweenness. Betweenness signifies how much an individual serves as a bridge between separate groups within their social networks. To compute betweenness, the shortest paths between every pair of individuals within a network are computed and the proportion of those paths that pass through a given individual contributes to their betweenness score. For example, imagine a classroom where Child A is the only child who has friends in two separate groups: any

shortest path between a child in one group and a child in the other requires going through A. Hence, Child A would have high betweenness because they lie on the shortest paths connecting many dyads. Individuals high in betweenness are believed to serve as important brokers or bridges in their networks, while individuals low in betweenness may be more socially peripheral (Prochnow, 2025). High gender crossers were significantly higher in betweenness than others ($M_{\text{high}} = 216.61$, $M_{\text{other}} = 100.25$), $t(236.07) = -9.12$, $p < .001$, 95% CI [-141.50, -91.23], suggesting that they represent bridges between clusters within classrooms (likely between clusters of mostly boys or mostly girls). High religion crossers, on the other hand, did not have significantly different levels of betweenness than other children ($M_{\text{high}} = 129.68$, $M_{\text{other}} = 116.33$), $t(538.32) = -1.58$, $p = .115$, 95% CI [-29.94, 3.24].

In summary, children who crossed gender lines for friendship and children who crossed religious lines for friendship seemingly held different social positions within their social networks. While both subgroups of children nominated more friends (and were perhaps more socially outgoing), high gender crossers appeared to be more popular among their peers (indexed by higher numbers of indegree nominations and closeness) and were more likely to act as bridges between clusters within their classroom (indexed by higher betweenness) than other children, High religion crossers similarly nominated significantly more than their peers but did not receive more nomination. Thus, even though cross-gender friendships were more rare than cross-religion friendships in our dataset, crossing gender lines was associated with more positive social outcomes than crossing religious lines. Even with these potential positive outcomes, however, both cross-religion and cross-gender friendships tended to be less reciprocated and hold less local transitivity, suggesting that these friendships were more structurally fragile.